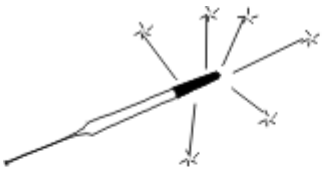
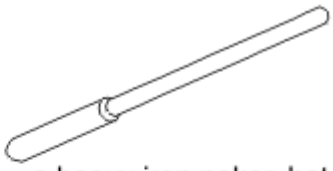
Q1.



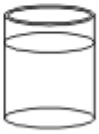
a tiny white hot spark from a sparkler



a heavy iron poker, hot enough to give out a dull red glow



a freshly boiled egg



a glass of cold water in a fridge

(a) Each of the four objects shown above started at room temperature. Now they are all at different temperatures, as described by the labels.

(i) Which object is at the highest temperature?

.....

1 mark

(ii) Which object has lost thermal energy?

.....

1 mark

(iii) Which object has had the largest gain in thermal energy?

.....

1 mark

(b) Omar puts a hot steel ball into a beaker of cold water. He waits until the temperatures of the ball and the water are the same.

From his results, he calculates that:

the thermal energy stored in the steel ball has fallen by 3770 J;
the thermal energy stored in the water has risen by 2940 J.

The energy stored in the water rose by less than 3770 J. What has happened to the other 830 J?

.....

.....

1 mark
Maximum 4 marks

Mark schemes

Q1.

(a) (i) the spark

accept 'the sparkler'

1 (L7)

(ii) the glass of water

accept 'the glass' ***or*** *'the water'*

1 (L7)

(iii) the poker

1 (L7)

(b) any **one** from

- energy transferred to beaker **or** air **or** surroundings

  accept 'thermal energy is lost to the beaker ***or*** *air* ***or*** *room'* ***or*** *'it has been dissipated'*

- beaker **or** air has been warmed up

- surroundings have warmed up

  accept 'used **or** *lost during evaporation'* **or** *'used to form steam'*

  accept 'lost as heat' **or** *'lost as sound'*

  do ***not*** *accept 'lost'*

  do ***not*** *accept 'evaporation'*

1 (L7)

[4]